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The harmonic resolvent operator can expose the fundamental energy-transfer pathways in fluid flows

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Fluid flows redistribute energy across multiple temporal scales through triadic frequency interactions arising from the quadratic advection term in the Navier-Stokes equations. More specifically, energy can be transferred from some source frequency α to a recipient frequency β through nonlinear mediation with frequency $\beta-\alpha$. These transfer mechanisms can sometimes be revealed through the analysis of simulation data, but collecting high-fidelity trajectories is often expensive and the results are inherently trajectory-dependent. In this paper, we show that when the flow is oscillatory and evolves in the proximity of time-periodic orbits (e.g., in wake flows, turbulent boundary layers, and several other configurations), these energy-transfer mechanisms can be studied through the lens of the harmonic resolvent, which is a frequency-domain input-output operator that inherently captures the leading-order cross-frequency interactions in the flow. Additionally, we establish that if the harmonic resolvent is approximately rank-1 (as is often the case for high-shear flows, especially in the proximity of bifurcation points), the operator's first left singular vector defines the *fundamental* mode-to-mode energy transfer tensor, which is an object describing the directed network that is preferentially used by the flow to exchange energy. We emphasize that this tensor can be computed exclusively with knowledge of the harmonic resolvent operator's singular vectors, without requiring simulation data. As a demonstrative example, we deploy these tools on the flow past an airfoil at Reynolds number 200 and angle of attack 20°.

Key words:

1. Introduction

In incompressible flows, energy transfer across different spatio-temporal scales is promoted by the quadratic advective term in the Navier-Stokes equations, which supports triadic frequency interactions between a source frequency α, a recipient frequency β and

a mediator frequency $\beta - \alpha$. Among other things, these interactions are responsible for the energy cascade in turbulent flows, the emergence of self-sustaining oscillations in wake flows, and the generation of higher harmonics in the response of a flow to periodic forcing. Revealing their nature and underlying mechanisms is therefore essential to improve our understanding of the flow physics and, eventually, control the flow behavior.

Currently, there exist several data-driven techniques to characterize energy interactions among different spatio-temporal scales. Bispectral Mode Decomposition (Schmidt 2020), for instance, identifies strongly coupled frequency combinations by maximizing the integral measures of bispectral density. Using data, this method identifies cascading harmonic interactions and their spatial support in physical space. However, the quantity of interest is not the energy-transfer rate, but rather an identification of coherent quadratic couplings of various modes. The energetic exchanges of coherent motion are further studied by Kinjangi & Foti (2023), where Dynamic Mode Decomposition is used to derive frequency-specific coherent kinetic energy budgets, quantifying energy exchange between coherent, mean, and random scales. Within this formulation, nonlinear inter-scale energy transfer is defined by a frequency triad of scale-specific velocities, however, it is not mode-to-mode. A related frequency-domain description is given by Triadic Orthogonal Decomposition (Yeung *et al.* 2026), which distinguishes donor, recipient, and catalyst frequencies for each triadic interaction and uses the covariance between the nonlinear convective field and the recipient velocity field to identify coherent mode pairs. A modal energy budget is then created, quantifying donor–recipient energy transfer and the associated spatial structures. In the case of turbulent channel flow, Ding *et al.* (2025) analyze the nonlinear transfer between Fourier modes indexed by streamwise and spanwise wavenumbers. By expanding the nonlinear transfer term into its individual triadic interactions, they define a conservative mode-to-mode transfer quantity that is mediated by an advective mode. All these methods are data-driven, as they require pre-computed trajectory data, and the computed energy interactions are therefore inherently trajectory-dependent. It is therefore natural to ask whether similar information can be inferred via a data-free, operator-theoretic approach.

Operator-theoretic approaches to energy transfer mechanisms describe the response of a flow directly from the governing equations, or their linearization around base flows of interest. For the case of a periodic cylinder wake, Jin *et al.* (2021) compare harmonic energy budgets obtained from direct numerical simulation (DNS) with those associated with the leading mean-flow resolvent modes. Their work shows that while resolvent analysis is able to reproduce the production and dissipation energy terms, it fails to reproduce the spatial distribution of nonlinear energy transfer along with the net nonlinear energy transfer into each harmonic. A connection between individual triadic interactions and the resulting flow response is established by Huang *et al.* (2026), who apply the mean-flow resolvent at each frequency-wavenumber pair to nonlinear forcing contributions extracted from the DNS of a turbulent channel flow. Their analysis quantifies how each triadic interaction contributes to the spectral kinetic energy of the response.

In this paper, we approach the analysis of energy-transfer mechanisms through the lens of the harmonic resolvent operator (Padovan *et al.* 2020; Padovan & Rowley 2022), which is a frequency-domain input-output map capturing the amplification and cross-frequency coupling of perturbations evolving in the proximity of a time-periodic base flow. As such, this operator is particularly well-suited to study the energy-transfer mechanisms between different temporal harmonics. First, we define a source-recipient transfer map (analogous to that considered in Ding *et al.* (2025) in the context of spatial wavenumbers) for general quadratically nonlinear systems, and show that this map admits a mode-to-mode interpretation when the nonlinearity satisfies an antisymmetry condition that guarantees conservative energy exchange. Then, we establish that if the harmonic resolvent

is approximately rank-1 (as is often the case in high-shear flows and flows supporting secondary instabilities), the operator's first left singular vector defines the *fundamental* mode-to-mode energy transfer tensor, which is an object describing the directed network that is preferentially used by the flow to exchange energy. We emphasize that this tensor can be computed exclusively with knowledge of the harmonic resolvent operator's singular vectors, without requiring simulation data. Finally, we apply these tools on the flow past a NACA 0012 airfoil, and we assess the accuracy of the predictions against direct numerical simulation.

2. Mathematical Formulation

2.1. *Energy transfer in a quadratically nonlinear system*

Consider a quadratically nonlinear system of the form

$$\frac{d}{dt}\boldsymbol{q}(t) = \boldsymbol{A}\boldsymbol{q}(t) + \boldsymbol{B}(\boldsymbol{q}(t), \boldsymbol{q}(t)) + \boldsymbol{h}(t), \quad \boldsymbol{q}(0) = \boldsymbol{q}_0 \in \mathbb{R}^n, \tag{2.1}$$

where $\boldsymbol{q}(t)$ is the state vector and $\boldsymbol{h}(t)$ is an exogenous forcing input. In the context of fluid flows, this equation may be understood as the spatially-discretized Navier-Stokes equation, where the state contains the flow variables over a computational grid, and $\boldsymbol{h}$ may be a volumetric or boundary force. Without loss of generality, let us consider the following decomposition of the state

$$\boldsymbol{q}(t) = \boldsymbol{Q}(t) + \boldsymbol{q}'(t), \tag{2.2}$$

where $\boldsymbol{Q}(t)$ is a solution of (2.1) and $\boldsymbol{q}'(t)$ is a vector of perturbations. By substitution into (2.1), we can derive the equation governing the dynamics of the fluctuations $\boldsymbol{q}'(t)$:

$$\frac{d}{dt}\boldsymbol{q}'(t) = \boldsymbol{L}(\boldsymbol{Q}(t))\boldsymbol{q}'(t) + \boldsymbol{B}(\boldsymbol{q}'(t), \boldsymbol{q}'(t)) + \boldsymbol{h}'(t), \tag{2.3}$$

where

$$\boldsymbol{L}(\boldsymbol{Q}(t))\boldsymbol{q}'(t) \coloneqq \boldsymbol{A}\boldsymbol{q}'(t) + \boldsymbol{B}(\boldsymbol{Q}(t), \boldsymbol{q}'(t)) + \boldsymbol{B}(\boldsymbol{q}'(t), \boldsymbol{Q}(t)). \tag{2.4}$$

Let us now define the perturbation energy as $E' = \langle \boldsymbol{q}', \boldsymbol{q}' \rangle / 2$, where $\langle \cdot, \cdot \rangle$ denotes a suitable energy inner product on $\mathbb{R}^n$. Taking the time rate of change of E' and using (2.3), we obtain

$$\begin{aligned} \frac{d}{dt}E'(t) &= \frac{1}{2}\left(\left\langle \frac{d}{dt}\boldsymbol{q}'(t), \boldsymbol{q}'(t) \right\rangle + \left\langle \boldsymbol{q}'(t), \frac{d}{dt}\boldsymbol{q}'(t) \right\rangle\right) \\ &= \underbrace{\left\langle \boldsymbol{q}'(t), \boldsymbol{L}(\boldsymbol{Q}(t))\boldsymbol{q}'(t) \right\rangle}_{R(t)} + \underbrace{\left\langle \boldsymbol{q}'(t), \boldsymbol{B}(\boldsymbol{q}'(t), \boldsymbol{q}'(t)) \right\rangle}_{N(t)} + \underbrace{\left\langle \boldsymbol{q}'(t), \boldsymbol{h}'(t) \right\rangle}_{H(t)}. \end{aligned} \tag{2.5}$$

Here, we identify three separate terms of interest: R represents the rate of energy production/dissipation associated with the linear dynamics around the trajectory $\boldsymbol{Q}(t)$, N is the term describing the rate of work done by the nonlinearity, and H is the rate of work done by the exogenous forcing term. The nonlinear term N is the centerpiece of the manuscript and, for energy-conserving nonlinearities, it satisfies $N(t) = 0$. From a physical standpoint, this equality implies that the nonlinearity does not introduce energy into nor dissipate energy from the system; it merely redistributes energy across different spatio-temporal scales. It is well-known that this is the case in incompressible flows with appropriate (e.g., no flux or periodic) boundary conditions.

In this paper, we are interested in understanding the energy transfer mechanisms across different temporal scales in flows evolving in the proximity of time-periodic orbits $\boldsymbol{Q}(t) =$

$\boldsymbol{Q}(t+\Theta)$, and it is therefore appropriate to introduce the following Fourier ansatz,

$$\boldsymbol{Q}(t) = \sum_{k\in\mathbb{Z}} \boldsymbol{Q}_k e^{ik\omega t}, \quad \boldsymbol{q}'(t) = \sum_{k\in\mathbb{Z}} \boldsymbol{q}'_k e^{ik\omega t}, \tag{2.6}$$

where $\omega = 2\pi/\Theta$ is the fundamental frequency of oscillation, and $\mathbb{Z}$ is the set of integers. The objective is to express the energy transfer term N in terms of the Fourier coefficients of $\boldsymbol{q}'$, and ultimately define a quantity describing the transfer of energy from one harmonic to the other. Defining $\boldsymbol{g}' \coloneqq \boldsymbol{B}(\boldsymbol{q}', \boldsymbol{q}')$, which is itself a Θ-periodic function of time with a well-defined Fourier series expansion, the energy transfer term may be expressed as

$$N(t) = \langle \boldsymbol{q}'(t), \boldsymbol{g}'(t)\rangle = \sum_{\substack{j,l\in\mathbb{Z}\\ l-j\in\mathbb{N}}} \underbrace{\gamma_{l-j}\,\Re\left(\langle \boldsymbol{q}'_j, \boldsymbol{g}'_l\rangle_{\mathbb{C}} e^{i(l-j)\omega t}\right)}_{N_{j,l}(t)} \tag{2.7}$$

Here, $\Re$ denotes the real part of a complex scalar, $\langle\cdot,\cdot\rangle_{\mathbb{C}}$ denotes the counterpart of $\langle\cdot,\cdot\rangle$ on $\mathbb{C}^n$, $\mathbb{N}$ is the set of natural numbers inclusive of 0, and $\gamma_{l-j} = 1$ if $l-j = 0$ and $\gamma_{l-j} = 2$ otherwise. In order to recover a physically-meaningful quantity describing energy transfer across different frequencies, we can manipulate $N_{j,l}(t)$ in equation (2.7) into

$$N_{j,l}(t) = \sum_{p\in\mathbb{Z}} \gamma_{l-j}\,\Re\left(\left\langle \boldsymbol{q}'_j, \boldsymbol{B}\left(\boldsymbol{q}'_{l-p}, \boldsymbol{q}'_p\right)\right\rangle_{\mathbb{C}} e^{i(l-j)\omega t}\right), \tag{2.8}$$

where we have used the definition of $\boldsymbol{g}'$ provided above and its Fourier coefficients. We then observe that for energy-conserving nonlinearities where $N(t) = 0$, we have

$$\frac{1}{\Theta}\int_0^\Theta N(t)\,dt = \sum_{j\in\mathbb{Z}} N_{j,j} = \sum_{j,p\in\mathbb{Z}} \underbrace{\Re\left\langle \boldsymbol{q}'_j, \boldsymbol{B}\left(\boldsymbol{q}'_{j-p}, \boldsymbol{q}'_p\right)\right\rangle_{\mathbb{C}}}_{\hat{T}_{(p,j-p)\to j}} = 0, \tag{2.9}$$

where the first equality follows from evaluating (2.7) with $j-l = 0$. The quantity $\hat{T}_{(p,j-p)\to j}$ may be interpreted as the *average* power supplied to (or extracted from) frequency $j\omega$ by the quadratic interaction of frequencies $p\omega$ and $(j-p)\omega$. We do remark, however, that by definition $\hat{T}_{(p,j-p)\to j}$ describes *triadic* interactions, and while it is clear that frequency $j\omega$ is the recipient frequency, it may not be immediately clear which, between $p\omega$ and $(j-p)\omega$, is the donor frequency. As discussed in Maeyama *et al.* (2021) and Ding *et al.* (2025) among others, $\hat{T}_{(p,j-p)\to j}$ may be understood as a *mode-to-mode* energy-transfer term between frequency pairs $(p\omega, j\omega)$ so long as the following equality is satisfied

$$\hat{T}_{(p,j-p)\to j} = -\hat{T}_{(j,p-j)\to p}. \tag{2.10}$$

This states that the energy received by frequency $j\omega$ from the quadratic interaction of $p\omega$ and $(j-p)\omega$ is equal and opposite to the energy received by $p\omega$ through the interaction of $j\omega$ and $(p-j)\omega$. Then, it follows that $p\omega$ is the donor in the triadic interaction $(p, j-p) \to j$, while $j-p$ is the mediator. It is important to observe that property (2.10) does not hold in general, and it is satisfied only under specific properties of the function $\boldsymbol{B}$, as established in the proposition below.

Proposition 1. *Equation* (2.10) *holds if* $\langle \boldsymbol{u}, \boldsymbol{B}(\boldsymbol{v},\boldsymbol{w})\rangle = -\langle \boldsymbol{w}, \boldsymbol{B}(\boldsymbol{v},\boldsymbol{u})\rangle$ *for any real-valued vectors* $\boldsymbol{u}$, $\boldsymbol{v}$ *and* $\boldsymbol{w}$.

Proof. Without loss of generality, assume that $\langle \boldsymbol{u},\boldsymbol{v}\rangle = \boldsymbol{u}^\top\boldsymbol{v}$ and $\langle \boldsymbol{a},\boldsymbol{b}\rangle_{\mathbb{C}} = \boldsymbol{a}^*\boldsymbol{b}$, where the asterisk denotes complex-conjugate transposition. (The proof for a general weighted inner product $\langle \boldsymbol{u},\boldsymbol{v}\rangle = \boldsymbol{u}^\top\boldsymbol{W}\boldsymbol{v}$ goes through analogously.) Since $\boldsymbol{B}(\boldsymbol{v},\boldsymbol{w})_i = \boldsymbol{H}_{ijk}\boldsymbol{v}_j\boldsymbol{w}_k$

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for some appropriate real-valued third-order tensor $\boldsymbol{H}$, it is straightforward to see that $\langle \boldsymbol{u}, \boldsymbol{B}(\boldsymbol{v},\boldsymbol{w})\rangle = -\langle \boldsymbol{w}, \boldsymbol{B}(\boldsymbol{v},\boldsymbol{u})\rangle$ implies $\boldsymbol{H}_{ijk} = -\boldsymbol{H}_{kji}$. Now, let $\boldsymbol{a}$, $\boldsymbol{b}$ and $\boldsymbol{c}$ be complex-valued vectors. Via direct manipulation, one can show that $\overline{\langle \boldsymbol{a}, \boldsymbol{B}(\boldsymbol{b},\boldsymbol{c})\rangle_{\mathbb{C}}} = -\langle \boldsymbol{c}, \boldsymbol{B}(\overline{\boldsymbol{b}},\boldsymbol{a})\rangle_{\mathbb{C}}$, from which it follows that $\Re\langle \boldsymbol{a}, \boldsymbol{B}(\boldsymbol{b},\boldsymbol{c})\rangle_{\mathbb{C}} = -\Re\langle \boldsymbol{c}, \boldsymbol{B}(\overline{\boldsymbol{b}},\boldsymbol{a})\rangle_{\mathbb{C}}$. Setting $\boldsymbol{a} = \boldsymbol{q}'_j$, $\boldsymbol{b} = \boldsymbol{q}'_{j-p}$ and $\boldsymbol{c} = \boldsymbol{q}'_p$, and using $\boldsymbol{q}'_{j-p} = \overline{\boldsymbol{q}'_{p-j}}$ since $\boldsymbol{q}'(t)$ is real-valued, equation (2.10) follows immediately, and this concludes the proof. □

The proposition essentially states that $\hat{T}_{(p,j-p)\to j}$ can be understood as a mode-to-mode energy-transfer term between frequency pairs $(p\omega, j\omega)$ if $\boldsymbol{B}$ satisfies a particular anti-symmetry property. Importantly, we shall see in the next subsection that the advective term $\boldsymbol{u}\cdot\nabla\boldsymbol{u}$ in the incompressible Navier-Stokes equation with no-flux or periodic boundary conditions enjoys this property, so the analysis of mode-to-mode energy transfer mechanisms in incompressible flows can be performed straightforwardly using $\hat{T}_{(p,j-p)\to j}$.

Remark 1. *We close this section by discussing an ambiguity that arises from the graph-theoretic interpretation of the energy transfer mechanisms. Specifically, let $\hat{S}_{p\to j} = -\hat{S}_{j\to p}$ denote any antisymmetric mode-to-mode energy transfer quantity that satisfies the total harmonic energy budget,*

$$\left(\boldsymbol{K}\hat{S}\right)_j \coloneqq \sum_{p\in\mathbb{Z}} \hat{S}_{p\to j} = \sum_{p\in\mathbb{Z}} \hat{T}_{(p,j-p)\to j} = N_{j,j}, \tag{2.11}$$

where $\boldsymbol{K}$ is the incidence (or discrete divergence) operator on the graph whose nodes are the harmonics $j\omega$. Since this graph has more edges than nodes, $\boldsymbol{K}$ has a non-trivial nullspace, whose elements are circulations on the graph. Consequently, the admissible transfers form an affine family $\hat{S} + ker(\boldsymbol{K})$ (where $ker(\boldsymbol{K})$ is the nullspace of $\boldsymbol{K}$), and $\hat{T}$ is one representative within it. As such, $\hat{T}$ may contain a circulating component that redistributes energy around closed loops of harmonics without altering the net energy received by any of them. While there has been extensive work (Dar et al. *2001; Maeyama* et al. *2021; Banerjee & Halder 2024; Singh* et al. *2025) focusing on pinning the gauge in the nullspace (e.g., via minimum-norm solutions corresponding to no circulation, or other constraints that uniquely determine the component of the solution along $ker(\boldsymbol{K})$), here we observe that $\hat{T}$ is not selected arbitrarily from the space of solutions of (2.11), but rather it comes from the governing equations and, as such, should be accepted as the physical solution of (2.11). Any cyclical component of $\hat{T}$ along the nullspace of $\boldsymbol{K}$ should therefore be accepted as physical.*

2.2. *Application to the incompressible Navier-Stokes equations*

Here, we consider the incompressible Navier-Stokes equation

$$\partial_t \boldsymbol{u} = -(\boldsymbol{u}\cdot\nabla)\boldsymbol{u} - \nabla p + \mathrm{Re}^{-1}\nabla^2\boldsymbol{u} + \boldsymbol{f}, \quad \nabla\cdot\boldsymbol{u} = 0, \tag{2.12}$$

over the spatial domain $\Omega \subseteq \mathbb{R}^3$ with no-flux boundary conditions $\boldsymbol{u}\cdot\boldsymbol{n} = 0$ at the domain boundary $\partial\Omega$. Here, $\boldsymbol{u}(\boldsymbol{x},t)$ is the three-dimensional vector velocity field, $p(\boldsymbol{x},t)$ is the pressure, and $\boldsymbol{f}(\boldsymbol{x},t)$ is an external forcing input. Shadowing the previous section, we decompose the velocity field and the pressure into a base flow and fluctuations (i.e., $\boldsymbol{u}(\boldsymbol{x},t) = \boldsymbol{U}(\boldsymbol{x},t) + \boldsymbol{u}'(\boldsymbol{x},t)$), and by substitution into (2.12), we obtain the evolution equation for the fluctuations

$$\partial_t \boldsymbol{u}' = -\boldsymbol{U}\cdot\nabla\boldsymbol{u}' - \boldsymbol{u}'\cdot\nabla\boldsymbol{U} + \mathrm{Re}^{-1}\nabla^2\boldsymbol{u}' - (\boldsymbol{u}'\cdot\nabla)\boldsymbol{u}' + \boldsymbol{f}' - \nabla p', \quad \nabla\cdot\boldsymbol{u}' = 0 \tag{2.13}$$

To draw a more straightforward analogy with the discussion in the previous section, it is useful to eliminate the pressure from the equation above using the Leray projector $\mathbb{P}$

onto the space of divergence-free vector fields. Specifically, with $\mathbb{P}\boldsymbol{v} \coloneqq \left(I - \nabla\Delta^{-1}\nabla\cdot\right)\boldsymbol{v}$, where Δ^{-1} is the inverse Laplacian with zero Neumann boundary conditions, the governing equation for the fluctuations reads

$$\partial_t \boldsymbol{u}' = \underbrace{\mathbb{P}\widetilde{\boldsymbol{L}}(\boldsymbol{U})\mathbb{P}\,\boldsymbol{u}'}_{\boldsymbol{L}} + \underbrace{\mathbb{P}\widetilde{\boldsymbol{B}}\left(\mathbb{P}\boldsymbol{u}', \mathbb{P}\boldsymbol{u}'\right)}_{\boldsymbol{B}(\boldsymbol{u}',\boldsymbol{u}')} + \underbrace{\mathbb{P}\boldsymbol{f}'}_{\boldsymbol{h}'} \tag{2.14}$$

where

$$\widetilde{L}(\boldsymbol{U})\boldsymbol{u}' = -\boldsymbol{U}\cdot\nabla\boldsymbol{u}' - \boldsymbol{u}'\cdot\nabla\boldsymbol{U} + \mathrm{Re}^{-1}\nabla^2\boldsymbol{u}', \quad \widetilde{\boldsymbol{B}}(\boldsymbol{u}', \boldsymbol{u}') = -\boldsymbol{u}'\cdot\nabla\boldsymbol{u}'. \tag{2.15}$$

We observe that (2.14) is now analogous to (2.3), as desired.

In order to define an energy-transfer quantity $\hat{T}$ as in the previous section, we must first verify that $\boldsymbol{B}$ is energy conserving in the sense of $N(t)$ (see equation (2.5)), and antisymmetric in the sense of Proposition 1. It suffices to show the antisymmetry condition $\langle \boldsymbol{u}, \boldsymbol{B}(\boldsymbol{v}, \boldsymbol{w})\rangle = -\langle \boldsymbol{w}, \boldsymbol{B}(\boldsymbol{v}, \boldsymbol{u})\rangle$, as the energy conservation property follows trivially by choosing $\boldsymbol{u} = \boldsymbol{v} = \boldsymbol{w}$. Specifically, given general vector fields $\boldsymbol{u}$ and $\boldsymbol{w}$ and a divergence-free vector field $\boldsymbol{v} = \mathbb{P}\boldsymbol{v}$ satisfying the no-flux boundary condition, we have

$$\langle \boldsymbol{u}, \widetilde{\boldsymbol{B}}(\boldsymbol{v}, \boldsymbol{w})\rangle = \int_\Omega v_j\partial_j\left(u_i w_i\right) - w_i v_j \partial_j u_i \, d\boldsymbol{x} = -\langle \boldsymbol{w}, \widetilde{\boldsymbol{B}}(\boldsymbol{v}, \boldsymbol{u})\rangle. \tag{2.16}$$

Here, we used integration by parts, as well as the divergence-free of $\boldsymbol{v}$ along with the no-flux boundary condition. The desired antisymmetry of $\boldsymbol{B}$ follows straightforwardly as shown below, where we use equation (2.16), the self-adjointness of $\mathbb{P}$ under the given inner product, and its idempotency,

$$\begin{aligned}\langle \boldsymbol{u}, \boldsymbol{B}(\boldsymbol{v}, \boldsymbol{w})\rangle &= \langle \boldsymbol{u}, \mathbb{P}\widetilde{\boldsymbol{B}}(\mathbb{P}\boldsymbol{v}, \mathbb{P}\boldsymbol{w})\rangle = \langle \mathbb{P}\boldsymbol{u}, \widetilde{\boldsymbol{B}}(\boldsymbol{v}, \mathbb{P}\boldsymbol{w})\rangle \\ &= -\langle \mathbb{P}\boldsymbol{w}, \widetilde{\boldsymbol{B}}(\boldsymbol{v}, \mathbb{P}\boldsymbol{u})\rangle = -\langle \boldsymbol{w}, \mathbb{P}\widetilde{\boldsymbol{B}}(\mathbb{P}\boldsymbol{v}, \mathbb{P}\boldsymbol{u})\rangle = -\langle \boldsymbol{w}, \boldsymbol{B}(\boldsymbol{v}, \boldsymbol{u})\rangle.\end{aligned} \tag{2.17}$$

Now that we have cast the Navier-Stokes equations into a form that is equivalent to (2.3), and verified that the Leray-projected advective term $\boldsymbol{B}$ is antisymmetric, the energy equation in Section 2.1 and the definition of $\hat{T}$ hold verbatim.

2.3. *Computing $\hat{T}$ via the harmonic resolvent*

In this section, we describe how $\hat{T}$ can be computed using the harmonic resolvent operator (Padovan *et al.* 2020; Padovan & Rowley 2022). We shall see that this approach is advantageous because, under mild assumptions, it allows us to analyze energy-transfer mechanisms in a flow without running computationally-expensive simulations.

We begin by assuming that $\boldsymbol{Q}(t) = \boldsymbol{Q}(t + \Theta)$ is a Θ-periodic solution of (2.3), and that $\boldsymbol{h}'(t)$ is a small-amplitude external forcing. Then, equation (2.3) can be approximated as a linear time-periodic dynamical system

$$\frac{d}{dt}\boldsymbol{q}'(t) = \boldsymbol{L}(\boldsymbol{Q}(t))\boldsymbol{q}'(t) + \boldsymbol{h}'(t), \tag{2.18}$$

where $\boldsymbol{L}(t) = \boldsymbol{L}(t + \Theta)$ is a Θ-periodic linear operator. Using the ansatz in (2.6), equation (2.18) may be expressed in the frequency domain as

$$\left[\boldsymbol{T}\hat{\boldsymbol{q}}'\right]_k \coloneqq ik\omega\hat{\boldsymbol{q}}'_k - \sum_{j\in\mathbb{Z}} \boldsymbol{L}_{k-j}\hat{\boldsymbol{q}}'_j = \hat{\boldsymbol{h}}'_k, \quad \forall k \in \mathbb{Z}, \tag{2.19}$$

where $\boldsymbol{L}_k$ denotes the kth Fourier coefficient of $\boldsymbol{L}$, $\hat{\boldsymbol{q}}' = \left(\ldots, \boldsymbol{q}'_{-1}, \boldsymbol{q}'_0, \boldsymbol{q}'_1, \ldots\right)$, and $\boldsymbol{T}$ is a harmonic-balanced, infinite-dimensional linear operator. The harmonic resolvent

operator $\boldsymbol{H}$ is then defined as the operator governing the long-term harmonic behavior of the linearized system (2.18) in response to harmonic forcing $\hat{\boldsymbol{q}}' = \boldsymbol{H}\hat{\boldsymbol{h}}'$, and it is closely related to the harmonic transfer function discussed in Wereley (1991). When the inverse of $\boldsymbol{T}$ exists, $\boldsymbol{H} = \boldsymbol{T}^{-1}$, otherwise, expressing $\boldsymbol{H}$ in terms of $\boldsymbol{T}$ requires some care. Specifically, it is well known from dynamical systems theory (Guckenheimer & Holmes 2002) that when $\boldsymbol{Q}(t+\tau)$ satisfies (2.1) with $\boldsymbol{h} = 0$ for any τ, then $d\boldsymbol{Q}(t)/dt$ is a solution of (2.18) with $\boldsymbol{h}' = 0$. Via the Fourier transform, it is straightforward to verify that the Fourier coefficients of $d\boldsymbol{Q}(t)/dt$ lie in the nullspace of $\boldsymbol{T}$ (i.e., $\boldsymbol{T}\widehat{d\boldsymbol{Q}/dt} = 0$), confirming that $\boldsymbol{T}$ is non-invertible. Here, we take the approach of Padovan *et al.* (2020), and define $\boldsymbol{H} = (\boldsymbol{T}|_{\Sigma})^{-1}$, where $\boldsymbol{T}|_{\Sigma}$ is the restriction of $\boldsymbol{T}$ to the subspace Σ orthogonal to $\widehat{d\boldsymbol{Q}/dt}$.

If we are interested in computing the energy-transfer mechanisms associated with (small-amplitude) flow fluctuations around a periodic orbit, then it becomes clear that the harmonic resolvent formulation can serve as a convenient approximation. Specifically, instead of time-stepping (2.3) under a prescribed external forcing input $\boldsymbol{h}'$, we can approximate the response $\hat{\boldsymbol{q}}'$ as $\hat{\boldsymbol{q}}' = \boldsymbol{H}\hat{\boldsymbol{f}}'$ and then substitute $\hat{\boldsymbol{q}}'$ directly into the definition of the mode-to-mode energy transfer term $\hat{T}$ in equation (2.9). For systems of moderate size (up to $O(10^6) - O(10^7)$ degrees of freedom at the time of writing), evaluating $\hat{\boldsymbol{q}}' = \boldsymbol{H}\hat{\boldsymbol{f}}'$ is often cheaper than time-stepping the dynamics in (2.3), especially if these are stiff. While potential computational savings are always welcome, the harmonic resolvent framework described herein also offers additional operator-theoretic insights that cannot be readily extracted from simulation data. In particular, suppose that $\boldsymbol{H}$ is (approximately) rank-1 with leading singular triplet $\left(\sigma_1, \hat{\boldsymbol{\phi}}_1, \hat{\boldsymbol{\psi}}_1\right)$, so that for any $\hat{\boldsymbol{h}}'$ we have

$$\hat{\boldsymbol{q}}' = \boldsymbol{H}\hat{\boldsymbol{h}}' \approx \sigma_1 \hat{\boldsymbol{\phi}}_1 \langle \hat{\boldsymbol{\psi}}_1, \hat{\boldsymbol{h}}' \rangle_{\mathbb{C}} = a_1 \hat{\boldsymbol{\phi}}_1, \quad a_1 \coloneqq \sigma_1 \langle \hat{\boldsymbol{\psi}}_1, \hat{\boldsymbol{h}}' \rangle_{\mathbb{C}} \in \mathbb{R}. \tag{2.20}$$

Then, by direct substitution of (2.20) into (2.9), we have

$$\Re\left\langle \boldsymbol{q}'_j, \boldsymbol{B}\left(\boldsymbol{q}'_{j-p}, \boldsymbol{q}'_p\right)\right\rangle_{\mathbb{C}} \approx a_1^3 \Re\left[\left\langle \hat{\boldsymbol{\phi}}_{1,j}, \boldsymbol{B}\left(\hat{\boldsymbol{\phi}}_{1,j-p}, \hat{\boldsymbol{\phi}}_{1,p}\right)\right\rangle_{\mathbb{C}}\right] \coloneqq a_1^3 \tilde{\hat{T}}_{(p,j-p)\to j}. \tag{2.21}$$

Remarkably, this means that up to a $\boldsymbol{h}'$-dependent scaling factor a_1^3, the energy transfer $\hat{T}$ may be expressed in terms of the first left singular vector of the harmonic resolvent operator. Thus, for an approximately rank-1 flow, the quantity $\tilde{\hat{T}}_{(p,j-p)\to j}$ serves as the *fundamental* mode-to-mode energy transfer tensor, describing how energy is transferred to/from different frequencies *for a general forcing* $\hat{\boldsymbol{h}}'$. We remark that the sign of $\tilde{\hat{T}}$ is arbitrary, since the singular vectors of an operator are unique up to a sign (after enforcing complex conjugacy). For a given external input $\hat{\boldsymbol{h}}'$, the sign is fixed by the scalar factor a_1^3.

3. Application to the flow over an airfoil

We study the energy-transfer mechanisms in the flow past a stationary NACA 0012 airfoil at angle of attack 20° and Reynolds number $Re = 200$, based on the freestream velocity u_∞ and chord length c. The incompressible Navier-Stokes equations are simulated using an in-house, spatially-second-order finite-volume solver implementing the immersed boundary via projection method described in Taira & Colonius (2007). The center chord of the airfoil is located at the origin of the spatial domain $\mathcal{X} = \mathcal{X}_y \times \mathcal{X}_x = [-4\,c, 4\,c] \times [-4\,c, 12\,c]$, discretized using a uniform, fully-staggered mesh with grid spacing $\Delta x = \Delta y = 0.02\,c$. At the left (inflow), top, and bottom boundaries, we impose Dirichlet velocity boundary conditions $(u, v) = (u_\infty, 0)$, and a zero-Neumann boundary condition at the right (outflow) boundary. We remark that these boundary conditions break the no-flux assumption of

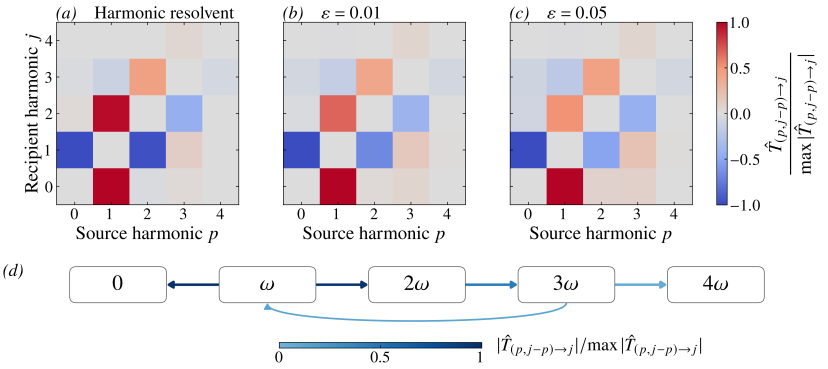


Figure 1. Comparison between the transfer map associated with the leading output mode of the harmonic resolvent (a) and the transfer maps reconstructed from snapshots from direct numerical simulation with $\varepsilon = 0.01$ (b) and $\varepsilon = 0.05$ (c). Directed network of inter-harmonic energy exchanges constructed from the transfer map $\hat{T}$ (d).

section 2.2. Thus, in all calculations involving (2.16), the non-vanishing surface integral is properly accounted for and subtracted out.

As discussed in Padovan *et al.* (2020), this flow configuration admits a polychromatic time-periodic solution with (non-dimensional) fundamental period $\Theta = 2\pi/\omega \approx 2.62$, computed by time-stepping into the post-transient regime. We take this solution as our base flow, truncate its frequency representation to $\Omega_b = \omega\{-3, \ldots, 0, \ldots, 3\}$, and consider fluctuations over the frequency set $\Omega = \omega\{-7, \ldots, 0, \ldots, 7\}$. We compute the SVD of the harmonic resolvent operator using the recently developed Python package `resolvent4py` (Padovan *et al.* 2025). As shown in figure 6a of Padovan *et al.* (2020), this operator is closely approximated as rank-1, with the first singular value σ_1 roughly 40× larger than the second one σ_2, justifying the use of the machinery developed in the previous section.

3.1. *Mode-to-mode energy transfer in the wake*

Using the first left singular vector of the harmonic resolvent, we compute the fundamental mode-to-mode energy transfer term $\tilde{\hat{T}}$ defined in (2.21). This is shown in figure 1a, where positive entries (red) denote energy transferred from the source harmonic p to the recipient harmonic j. The content of this figure is processed into a directed graph, shown in figure 1(d), showcasing the energy transfer cascade among harmonics of the fundamental vortex shedding frequency ω. These figures reveal that the most dominant energy transfer involves the fundamental frequency ω, which donates energy to its second harmonic 2ω and the mean 0. Additionally, frequency 2ω donates energy to 3ω, which, in turn, donates energy to 4ω, thus suggesting that the flow supports unidirectional energy transfer from low to high frequencies. We do recall, however, that the fundamental mode-to-mode energy transfer tensor discussed herein is unique up to a sign, so there can be external forcing vectors $\boldsymbol{h}'$ that trigger the opposite cascade, with energy flowing from high to low frequencies, and ω receiving energy from the mean. We also observe that 3ω donates a small amount of energy to ω, thus creating a circulation (see Remark 1 in section 2.1).

To test the accuracy of the energy transfer prediction put forth by the harmonic resolvent operator, we compare with the ground truth generated via numerical integration of the nonlinear Navier-Stokes equation. Specifically, we move the airfoil in the vertical direction with sinusoidal velocity $v = \varepsilon\omega\cos(\omega t)$ for $\varepsilon = 0.01$ and $\varepsilon = 0.05$, and integrate the equations to the post-transient (forced) limit cycle. Since, as discussed in section 2.3, the harmonic resolvent is defined up to a phase shift along the base flow, we phase match the

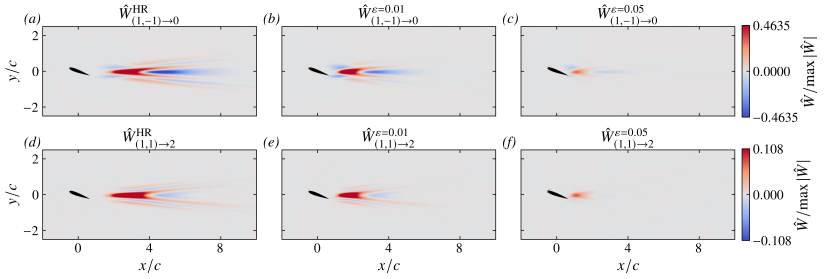


Figure 2. Comparison of the spatial distribution of nonlinear energy transfer corresponding to transfer map entries $\hat{T}_{(1,-1)\to 0}$ (a,b,c) and $\hat{T}_{(1,1)\to 2}$ (d,e,f) between computation by direct numerical simulation with varying ε and the rank-1 approximation.

post-transient response to the base flow in the least-squares sense as shown in equation (3.9) in Padovan *et al.* (2020). For smaller heaving motions ($\varepsilon = 0.01$), we expect to see good agreement since the flow should behave linearly, but we are also interested in understanding whether the harmonic resolvent prediction is qualitatively correct in more nonlinear regimes induced by a stronger airfoil motion ($\varepsilon = 0.05$). Figures 1(b,c) show the mode-to-mode energy transfer in the two forcing cases compared to the rank-1 prediction in figure 1a. All three maps identify the fundamental frequency as the principal energy donor, supplying the zero-frequency component and second harmonic. The subsequent transfers from 2ω to 3ω and from 3ω to 4ω, together with the weaker return from 3ω to ω, are also recovered. The persistence of these pathways across both forcing amplitudes shows that the leading harmonic-resolvent mode captures the dominant organization of nonlinear energy transfer, with the largest discrepancies confined to higher-order pathways whose agreement improves as the forcing amplitude decreases.

3.2. *Spatial structures associated with the transfer mechanisms in the wake*

The spatial structures associated with the energy transfer mechanisms discussed earlier can be readily computed from the work density

$$\hat{W}_{(p,j-p)\to j}(\boldsymbol{x}) = \Re\left\langle \boldsymbol{q}'_j(\boldsymbol{x}), \boldsymbol{B}\left(\boldsymbol{q}'_{j-p}(\boldsymbol{x}), \boldsymbol{q}'_p(\boldsymbol{x})\right)\right\rangle, \tag{3.1}$$

where $\langle\cdot,\cdot\rangle$ is the standard inner product on $\mathbb{C}^{\dim(\boldsymbol{x})}$ (here, $\mathbb{C}^2$). For clarity, we remark that $\hat{T}$ is recovered from $\hat{W}$ by integrating the latter over the volume. Here, we evaluate $\hat{W}$ for the two strongest energy transfers $\hat{T}_{(1,-1)\to 0}$ and $\hat{T}_{(1,1)\to 2}$, where ω is the donor and 0 and 2ω are the recipients, respectively. Figure 2 shows the fields corresponding to $\hat{W}$ as computed from DNS at the two different forcing amplitudes and from the rank-1 harmonic resolvent approximation. Figures 2(a,d) and 2(b,e) show good qualitative agreement between the harmonic resolvent prediction and the low-amplitude case, even though we observe that the latter reveals spatial structures that are closer to the trailing edge of the airfoil. By contrast, the spatial field associated with the higher-amplitude motion (Figure 2(c,f)) shows a much stronger qualitative departure from the harmonic resolvent prediction, and this is indicative of second-order nonlinear interactions that are not captured by the linearization.

4. Conclusion

We leverage the harmonic resolvent operator to describe the nonlinear transfer of energy between different temporal frequencies in incompressible fluid flows. After identifying a source-recipient transfer map describing the energy transfer pathways in energy-conserving

quadratic systems, we showed that this map can be well approximated using information collected from the harmonic resolvent. More specifically, we established that if the harmonic resolvent operator is (approximately) rank-1, then one can compute a *fundamental* (i.e., trajectory-independent) mode-to-mode transfer map describing the energy transfer pathways for any perturbed trajectory. We applied this framework to vortex shedding past an airfoil, whose harmonic resolvent exhibits approximate rank-1 structure, and found that energy flows primarily from the fundamental frequency to the mean and neighboring harmonics. Comparison with DNS of a sinusoidally heaving airfoil confirms that the approximation captures both the dominant source-recipient network as well as the dominant spatial regions of energy exchange.

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